



Mikhaïl **WATCHUENG TCHEKAM**

Biomedical Engineering

MY PROFILE

Specializing in biomechanics, medical robotics, and the analysis and modeling of image-based systems. Highly determined and autonomous, I have the ability to lead a complex technical project to completion, both independently and as part of a multidisciplinary team. Motivated to contribute to clinical research, medical imaging, RWD-based modelling, and robotics innovation.

CONTACT DETAILS

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PERSONAL INFORMATION

Citizenship: **Cameroonian**
Family: **Single, no children**
Languages: **French** (native),
English (B2), **Dutch** (A2), **German** (A2)

TECHNICAL SKILLS

- **Programming:** Python, C/C++, R, Matlab, Java, JavaScript
- **Data Science:** machine learning, data engineering
- **Imaging:** OCT basics, segmentation, 3D reconstruction principles
- **Robotics:** robot modelling, kinematics/dynamics (ROBOTRAN)
- **Tools:** SOLIDWORKS, ROS basics

EDUCATION

MASTER IN BIOMEDICAL ENGINEERING (Biomechanics & Medical Robotics), EPL — UCLouvain. **2026**

◊ Advanced courses in biomechanics, medical imaging, medical robotics, machine learning, multibody dynamics, and modelling of biological systems.

BACHELOR IN CIVIL ENGINEERING (Biomedical & Mechanical Engineering), EPL — UCLouvain. **2023**

◊ Projects in system design, mechanics, imaging analysis, and robotics control.

EXPERIENCE

RESEARCH INTERN (ROBOTICS) — *BruBotics*. **2025.09–Present**

◊ Working on OCT-based imaging analysis, 3D reconstruction, and tool–retina depth estimation using shadow-based methods. Contributing to robotic-assisted eye surgery research, integrating imaging and control concepts.

RESEARCH PROJECT MEMBER — *Neurotech, Leuven*. **2024–2025**

◊ Signal analysis, vibrotactile processing, EEG pattern exploration, Python-based pipelines.

JUNIOR MEMBER — *Orsi Academy*. **2024–2025**

◊ Hands-on exposure to surgical robotics, system modelling, and performance evaluation.

TUTOR (MATH, PHYSICS, CHEMISTRY, BIOLOGY) — *MySherpa & La Chaloupe*. **2016–Present**

◊ Development of analytical and pedagogical methods for teaching complex STEM concepts.

PRODUCTION & LOGISTICS WORKER. **2015–2023**

◊ Operational precision, workflow optimization, reliability in high-pressure environments.

SOFT SKILLS

Autonomy, rigour, teamwork, analytical thinking, rapid learning, precision, adaptability, scientific communication.

HOBBIES

Basketball, football, movies analysis.